Challenge 0

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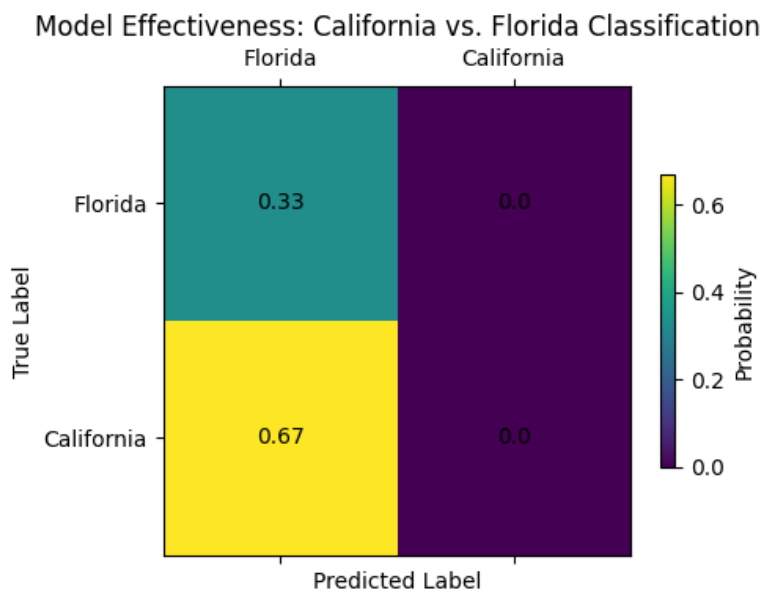
Objective

The goal is to implement a classification model by building "from scratch" the three main regularized logistic regression models discussed in class (Ridge, Lasso, and Elastic Net). The effectiveness of the models will then be evaluated using appropriate metrics. Finally, a function will be implemented to calculate the points needed to plot the ROC curves.

First model plot

The numerical probabilities have been added to the graph to make reading clearer.



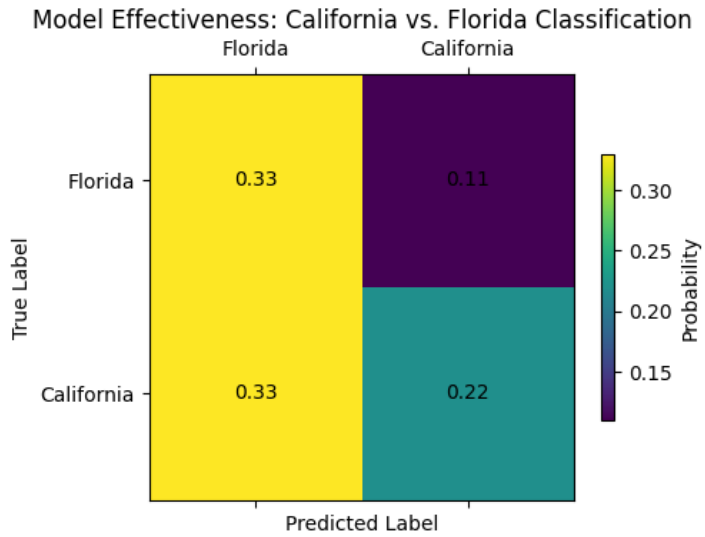
Comment on the chart

It seems the model has learned a "shortcut": always predict "Florida". This result suggests two things:

1. The model likely saw training data where "Florida" was the most common or "safest" answer to minimize error.
2. The model failed completely, collapsing onto a single class and achieving very poor performance on the test set (3 out of 9 correct).

Solution

It was decided to make the data division less "unlucky" by forcing `train_test_split` to maintain the same percentage of "Florida" and "California" (the original proportion) in both the training and test sets. The `stratify` method was therefore used.



Results Commentary

The model is clearly "better" (or less "poor") at recognizing "*Florida*" (**33%** correct predictions) compared to "*California*" (**22%** correct predictions). Crucially, it makes significantly more *serious* errors when it encounters a Californian startup (making a mistake in 33% of cases) compared to when it encounters one from Florida (making a mistake in only 11% of cases).

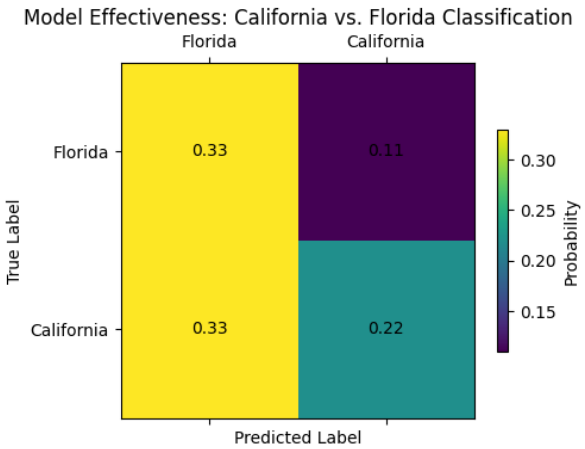
Implementation of Regularized Models

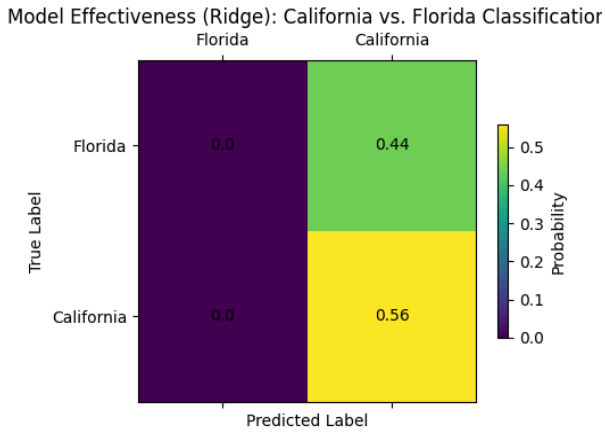
The parameters used in the implementation of the models are the following:

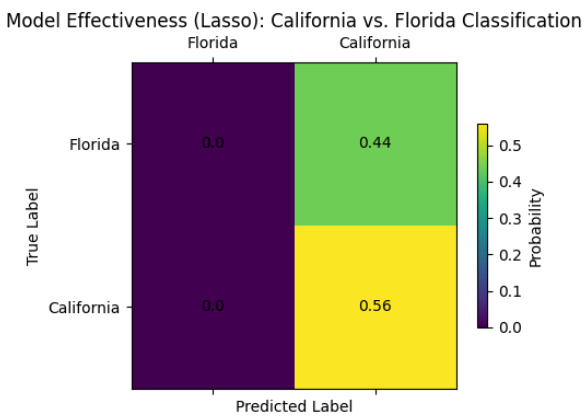
Number of iterations	10.000	2nd Lambda (Elastic Net)	0.5
Learning rate	1e-9	Threshold	0.5
Lambda	0.1		

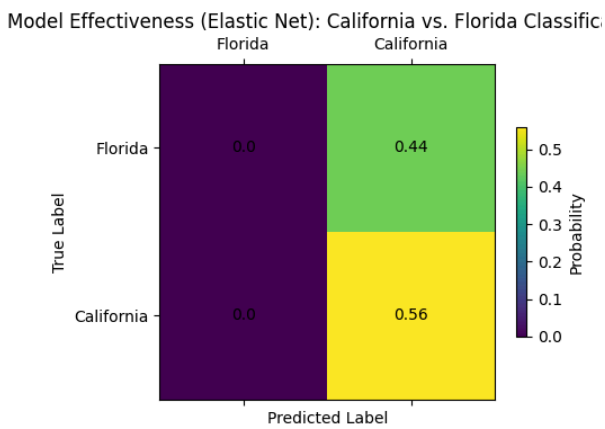
Plot and Results Commentary

From the confusion matrices, it is evident that the models tend to "learn" only one class (California) and consequently "guess" it with a percentage close to 50%; therefore they are not actually learning but here too they have found a shortcut.









The **classification report** highlights a severe failure in classifying the "Florida" class across all three models tested, with Precision, Recall, and F1-score all at 0.0. This means that **no model was able to correctly identify a single sample of the "Florida" class**. Probable causes include:

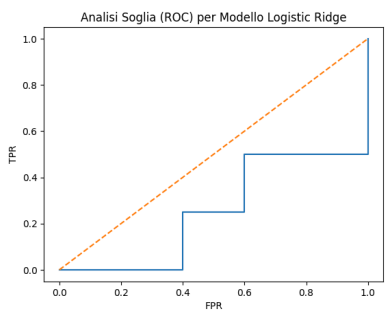
1. **Insufficient Sample:** The "Florida" class has too few data points for learning.
2. **Optimization Issues:** Poor optimization of Gradient Descent algorithms, with models "stuck" in a local minimum/saddle point.
3. **Identical Results:** The models collapse onto the same trivial solution (likely null coefficients).

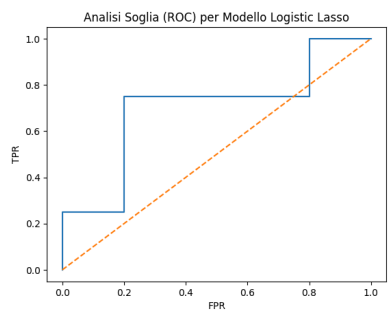
Conclusion: The 55.6% **accuracy** is **misleading** and represents only the **baseline**, which is the accuracy achieved by always guessing the most numerous class ("California"). The models have no predictive power.

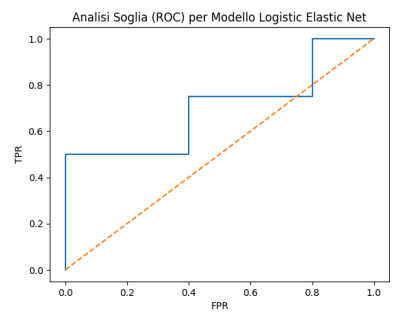
ROC implementation

A ROC function has been implemented that takes the probabilities calculated by the various models and uses them cyclically as thresholds to then calculate the **TPR** and **FPR**. These values are progressively saved in 2 arrays and then the points are plotted to visualize the **ROC** curve.

Plot and Results Commentary







The Lasso and Elastic Net models have a weak but real predictive power (for *ranking*), which was masked by a very poor threshold calibration (the 0.5). The Ridge model, on the other hand, is a total failure, worse than the random case.

Lasso and Elastic Net (AUC ≈ 0.70 - 0.75):

- **Result:** Very good, clearly better than the random case (AUC > 0.5).
- **ROC Behavior:** The curve rises vertically at the beginning (excellent sign), indicating that high-probability predictions were correct (true positives).
- **Problem:** The failure in the *classification report* is due to a **poor calibration** (probabilities all < 0.5), not a lack of *ranking* power. Choosing a different optimal threshold (e.g., 0.2) would make them predictive.

Ridge (AUC ≈ 0.25 - Disastrous):

- **Result:** Complete failure (AUC < 0.5), worse than the random case.
- **ROC Behavior:** The curve initially shifts horizontally, meaning that the most "certain" predictions (highest probabilities) were actually False Positives.
- **Problem:** The model learned the correlation **in reverse**, assigning higher probabilities to negatives than to positives. The ranking is inverted and harmful.